

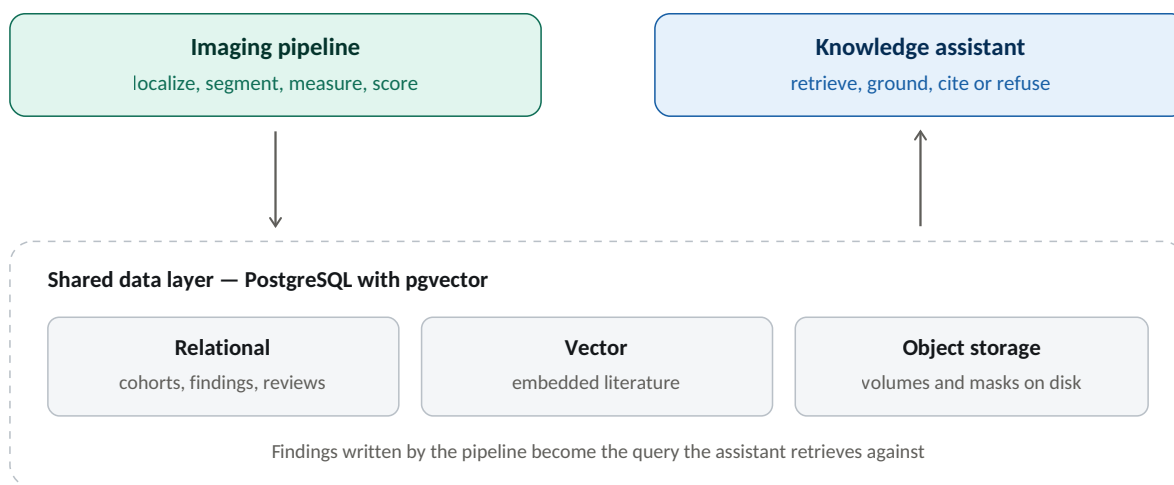
Autonomous Pancreatic Lesion Detection and Radiologist Review System

Quinn Evans · Supplement to the capstone proposal

This appendix documents the system design behind the proposal. It is included so a reviewer can assess whether the architecture has been thought through, not only whether the technologies have been named. Every quantity given here was measured against the real dataset or taken from published documentation; estimates are labeled as such.

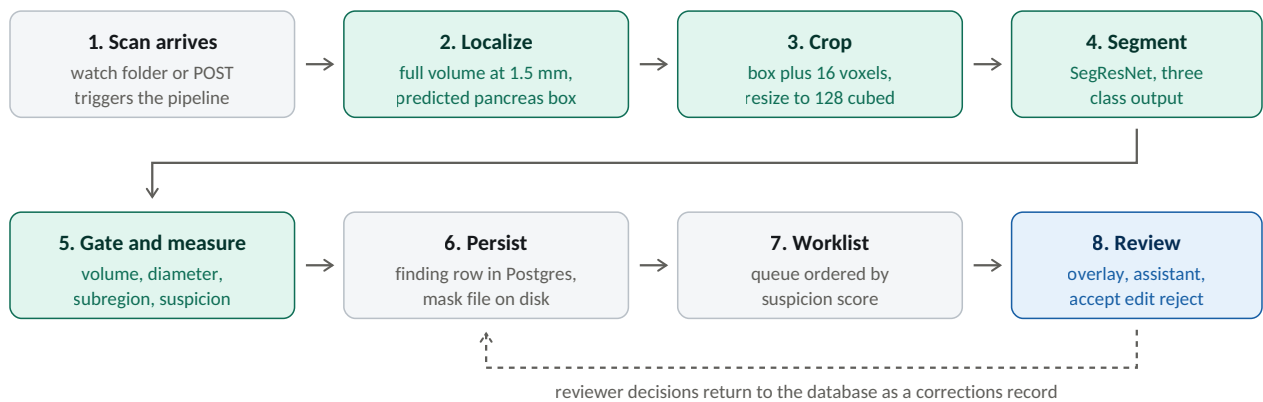
A1. System architecture

The project consists of two systems that meet at a shared data layer. The imaging pipeline produces structured findings; those findings become the retrieval context for the knowledge assistant. Neither system calls the other directly.



A2. End-to-end data flow

The following is the complete path a scan takes. Steps 2 through 5 are the autonomous cascade that this capstone builds; today step 2 is replaced by a human-provided region of interest.



Measured inference cost. A full CT resampled to 1.5 mm isotropic is a median of $253 \times 190 \times 142$ voxels — 6.9 million voxels, 26.5 MiB as float32, computed across all 9,901 scans in the dataset. Sliding-window localization at 128^3 with 50 percent overlap requires a median of 12 windows per scan. Against the current serving time of roughly 0.6 seconds for the provided-region path, autonomous whole-scan inference is expected to land in the range of several seconds per study.

A3. Relational schema

Abbreviated to the tables that carry the design decisions. The full schema adds audit columns and indexes.

```

-- Storage roots are configuration, never absolute paths in data.
CREATE TABLE storage_root (
  id          SERIAL PRIMARY KEY,
  name        TEXT NOT NULL UNIQUE,      -- 'cold_archive', 'warm_cache'
  tier         TEXT NOT NULL CHECK (tier IN ('cold', 'warm', 'cloud'))
);

CREATE TABLE scan (
  id          BIGSERIAL PRIMARY KEY,
  case_id     TEXT NOT NULL,
  source_dataset TEXT NOT NULL,          -- 'pants' | 'panorama'
  shape       INT[3] NOT NULL,
  spacing     REAL[3] NOT NULL,
  contrast_phase TEXT,
  manufacturer TEXT,
  site        TEXT,
  UNIQUE (source_dataset, case_id)
);

-- Provenance is data, not an assumption. PANORAMA organ masks are machine
-- generated; PANTS masks are expert validated. The model must know which.
CREATE TABLE annotation (
  id          BIGSERIAL PRIMARY KEY,
  scan_id     BIGINT NOT NULL REFERENCES scan(id),
  structure    TEXT NOT NULL,           -- 'pancreas' | 'lesion' | ...
  provenance   TEXT NOT NULL CHECK (provenance IN ('expert', 'automated')),
  volume_mm3   DOUBLE PRECISION
);

```

```

CREATE TABLE scan_file (
    id          BIGSERIAL PRIMARY KEY,
    scan_id     BIGINT NOT NULL REFERENCES scan(id),
    storage_root_id INT NOT NULL REFERENCES storage_root(id),
    relative_path TEXT NOT NULL,
    bytes       BIGINT,
    sha256      CHAR(64),
    verified_at TIMESTAMPTZ
);

-- A cohort is a materialized query, not a text file. This is the control that
-- makes the validation leak found in the preceding project inexpressible.
CREATE TABLE cohort (
    id          SERIAL PRIMARY KEY,
    name        TEXT NOT NULL UNIQUE,
    definition   TEXT NOT NULL,          -- the SQL that produced it
    parent_fold TEXT,                  -- 'train' | 'val' | 'test'
    frozen_at   TIMESTAMPTZ
);

CREATE TABLE cohort_member (
    cohort_id INT NOT NULL REFERENCES cohort(id),
    scan_id   BIGINT NOT NULL REFERENCES scan(id),
    PRIMARY KEY (cohort_id, scan_id)
);

-- Enforced at write time: a scan may never appear in a training cohort and an
-- evaluation cohort simultaneously.
CREATE OR REPLACE FUNCTION assert_cohort_disjoint() RETURNS TRIGGER AS $$
BEGIN
    IF EXISTS (
        SELECT 1 FROM cohort_member cm
        JOIN cohort c  ON c.id = cm.cohort_id
        JOIN cohort nc ON nc.id = NEW.cohort_id
        WHERE cm.scan_id = NEW.scan_id
        AND c.parent_fold IS DISTINCT FROM nc.parent_fold
    ) THEN
        RAISE EXCEPTION 'scan % would cross folds', NEW.scan_id;
    END IF;
    RETURN NEW;
END; $$ LANGUAGE plpgsql;

CREATE TABLE prediction (
    id          BIGSERIAL PRIMARY KEY,
    scan_id     BIGINT NOT NULL REFERENCES scan(id),
    model_version TEXT NOT NULL,
    lesion_volume_mm3 DOUBLE PRECISION,
    max_diameter_mm  DOUBLE PRECISION,
    subregion        TEXT,
    suspicion_score   DOUBLE PRECISION,    -- drives worklist ordering
    mask_file_id     BIGINT REFERENCES scan_file(id),
    UNIQUE (scan_id, model_version)
);

CREATE TABLE review (
    id          BIGSERIAL PRIMARY KEY,
    prediction_id BIGINT NOT NULL REFERENCES prediction(id),
    decision     TEXT NOT NULL CHECK (decision IN ('accept','edit','reject')),
    reviewer     TEXT NOT NULL,
    decided_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
    note         TEXT
);

-- Literature chunks share the database with the imaging metadata, so a single

```

```
-- query can combine semantic similarity with structured filters.
CREATE TABLE doc_chunk (
  id          BIGSERIAL PRIMARY KEY,
  pmid        TEXT NOT NULL,
  title       TEXT,
  journal     TEXT,
  year        INT,
  chunk_text  TEXT NOT NULL,
  embedding   VECTOR(768)
);
CREATE INDEX ON doc_chunk USING hnsf (embedding vector_cosine_ops);
```

Why this matters. In the preceding project, cohort membership lived in text files, and a script sampled from the wrong column. The result was 266 evaluation cases inside the training set and a headline lesion Dice inflated from 0.415 to 0.528. That number had to be withdrawn. Expressing membership as a constrained relation makes the same mistake impossible to commit rather than merely possible to detect.

A4. Cross-source integration

PanTS and PANORAMA describe the same anatomy under incompatible conventions.

	PanTS	PANORAMA
label values	1 = pancreas, 2 = lesion	1 = PDAC, 2 = veins, 3 = arteries, 4 = parenchyma, 5 = duct, 6 = bile duct
lesion annotation	expert validated	482 expert, 194 automated
organ annotation	expert validated	automated throughout
negative standard	no lesion mask present	histopathology or 36 month follow-up
tumor types	PDAC, IPMN, PNET	PDAC only
contrast phase	four phases	portal venous only

Deduplication. PANORAMA states that its public set includes 194 Medical Segmentation Decathlon cases and 80 NIH cases. Those sources are already inside PanTS, which contains 966 single-institution Memorial Sloan Kettering cases and 97 NIH cases. The overlap was confirmed by two independent signals: institution matching against the site metadata, and a tumor-volume distribution fingerprint — PanTS and the Decathlon share a single 732 cm³ tumor, an outlier neither dataset has twice.

	cases	tumor positive
PANORAMA public set	2,238	676
less Decathlon contribution	-194	-98
less NIH contribution	-80	0
usable, non-overlapping	1,964	578

Why this is load-bearing. The training fold contains 706 tumor-positive scans and the preceding project already used all 706. Scaling within PanTS therefore adds only healthy scans, which was tested and rejected — specificity rose from 17 to 46 percent but detection fell from 96 to 88 percent, below the pre-registered floor. PANORAMA is the only available source of additional tumors, and the merge cannot proceed safely until deduplication and remapping are correct.

Residual risk, disclosed. 533 PanTS records name Radboud inside a pooled multi-institution provenance string. No single-institution Dutch case exists in PanTS, but without case-level identifiers a small overlap cannot be fully excluded. This will be stated in the results rather than claimed away.

A5. Storage tiers and compute

Measured across all 9,901 PanTS scans: 297.5 billion voxels total.

tier	contents	size	location
cold	original compressed volumes, both datasets	582 GB	external drive, checksummed, never read during training
warm	preprocessed 128 ³ arrays, 10.0 MiB per case	25 GiB for the merged training cohort	internal SSD or cloud volume
index	metadata, cohorts, findings, embeddings	~1–2 GB	PostgreSQL

Uncompressed, the imaging archive would occupy 831 GB, and storing all 28 annotated structures as separate full volumes would require 8.1 TB. Neither is necessary: training reads only the warm tier.

Consequence for compute. Moving training to a rented GPU requires uploading 25 GiB once, not 582 GB. Three controls preserve comparability with existing results: numerical precision stays at fp32, the baseline is re-established on the new hardware before any comparison is drawn, and on-demand rather than preemptible instances are used until checkpoint resume is verified end to end.

A6. Knowledge assistant

Corpus. PubMed abstracts and the PMC Open Access subset, filtered by MeSH terms covering pancreatic neoplasms and computed tomography. Estimated 30,000–80,000 articles yielding roughly 300,000–500,000 chunks; both figures to be confirmed at build time. The snapshot is pinned by date and query, because a corpus that changes underneath the system makes retrieval metrics from week 6 incomparable to week 10.

Pipeline.

stage	specification
chunking	approximately 512 tokens, 64 token overlap, section boundaries respected
embedding	sentence-transformer; pre-registered comparison of a general model (384 dimensions) against a biomedical model (768 dimensions)
index	pgvector HNSW, cosine distance, parameters tuned against a held-out question set
retrieval	top-k passages, filtered by structured predicates in the same SQL query
generation	Qwen instruct via Ollama, temperature 0, executed locally

Query construction. The assistant is not a free-text chatbot. A query is templated from the finding row the pipeline just wrote:

```
finding: lesion 2.4 cm maximum diameter, 5.3 cm3, pancreatic head,
        portal venous phase, organ volume 77 cm3
question: <clinician question>
retrieve: top-k chunks WHERE embedding <=> query_embedding
        AND year >= 2010
```

Cite or refuse. The system prompt permits only statements supported by the retrieved passages, each rendered in the interface beside the text it supports. If the best retrieval score falls below a calibrated threshold, the assistant returns a refusal rather than generating. This is a testable property, and it is what keeps a literature-lookup tool from becoming unlicensed clinical advice.

Evaluation. Retrieval is scored with recall@k and mean reciprocal rank against a hand-authored question set with known-relevant documents. Generation is scored by groundedness rate — the fraction of generated claims traceable to a retrieved passage — and by refusal accuracy on questions deliberately outside the corpus.

A7. Risks and fallbacks

risk	likelihood	response
The Week 4 experiment returns null — added tumors do not improve lesion Dice	Moderate	Reported as a null result and the baseline cohort is retained. The integration work stands on its own as the data engineering deliverable; the pipeline and gate are unaffected.
The autonomous cascade loses significant accuracy against the provided-region baseline	Moderate	Both numbers are reported side by side, which is the honest comparison in any case. Prior work measured 98 percent tumor coverage from the predicted region, so a total failure is unlikely.
Johns Hopkins does not respond to the external submission	Likely enough to plan for	The submission is a stretch goal outside the critical path. Two alternative external validations remain: leave-one-institution-out across six sites within PanTS, and the PANORAMA challenge's own hidden test cohort.
Specificity cannot be raised without breaching the 96 percent detection floor	Moderate	Report the full operating-point curve rather than a single threshold, and state plainly which trade-offs are available. A prior experiment already established that this axis is movable and located the cost in small tumors.
PANORAMA download or licensing proves impractical	Low	CC BY-NC 4.0 is confirmed and the data is on Zenodo. If it fails, the project continues on PanTS alone with a reduced Week 4 scope.
Assistant produces an ungrounded clinical statement	Low, and gated	Cite-or-refuse is enforced at generation, not by disclaimer, and refusal accuracy is measured. Any ungrounded output reaching the interface is treated as a defect, not a tuning issue.
Training throughput on local hardware is insufficient	Moderate	The warm cache makes cloud GPU a 25 GiB upload. A profiling pass in Week 1 determines whether the bottleneck is compute or input/output before any money is spent.
Scope overruns across two systems	Moderate	Each pillar carries a committed minimum and an optional stronger version, so a difficult week costs sophistication rather than a component.

A8. What already exists

The following is complete and working before Week 1, and is the foundation the capstone builds on.

- A registered model in MLflow with version, step, and checksum, trained by transfer learning from the SuPreM pretrained checkpoint.
- Evaluation on the official 901-scan held-out test set: lesion Dice 0.474 with a 95 percent confidence interval of 0.42 to 0.52, pancreas Dice 0.827, detection sensitivity 96 percent, specificity 17 percent, patient-level AUC 0.804. These are provided-region results, and are reported as such.
- A FastAPI inference service and a React and NiiVue review interface with prediction overlays and 3D reconstruction.
- Twenty-seven pre-registered experiments with recorded accept or reject decisions, including two rejections and one withdrawn result.

- A localize-then-segment cascade that runs end to end and has been measured once at 98 percent tumor coverage from the predicted region.
- Two independent adversarial code audits, one of which found the validation leak described in A3.

The capstone is not a rebuild. It removes the system's central assumption, adds a second data source and a second AI system, and replaces convention with enforcement in the data layer.